

COMMENT

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In trying to understand and interpret the results of this paper, the one point which gave me the greatest difficulty was that of the possibility of negative information from the data. The authors give a correct *technical* explanation in terms of discrepancy between the prior and the sample information; but this still leaves philosophical and practical questions unanswered. In the life testing problem, if I observe $\bar{x} < 500$ or $\bar{x} > 8000$, am I really worse off than without the sample? Surely not! To resolve this issue, we need to look carefully at what their $I_{\bar{x}}$ really measures: it seems to me that it measures the reduction in entropy (=chaos), i.e. the reduction in confusion in the mind of the observer (analyst/decision maker/...). It is common experience that new data can muddy rather than clarify (initially at least) the situation, i.e. confusion is increased, or $I_{\bar{x}} < 0$. But is it correct to blame the data for this increased confusion, by saying that the data contribute negative information? The data *do* contain very real information, even if it is the unwelcome news that our specified prior may have claimed greater confidence than justified.

With the above in mind, it seems that a better measure of the *new* information contributed by the data, relative to the prior, is the degree of shift from prior to posterior occasioned by the data. The Kullback-Leibler measure, given by $J_{\bar{x}}$ in equation (1.9) of the paper, measures precisely this shift. We can then view the two measures in the following terms:

$I_{\bar{x}}$: The increase in the observer's *perceived* state of information (= the reduction of confusion in the mind of the observer);

$J_{\bar{x}}$: The amount of *new* information contributed by the data.

As the authors point out, these two quantities have the same expectation, a philosophically interesting point which merits further consideration. The authors examine only the behaviour of $I_{\bar{x}}$ as $\bar{x}$ varies. I briefly looked at what happens to $J_{\bar{x}}$ as $\bar{x}$ varies in their life-testing example, with two observations. This is illustrated in Figure C1. The contrast with Figure 1 in the paper is evident: whereas $I_{\bar{x}}$ has a maximum near the a priori expectation of $\bar{x}$, $J_{\bar{x}}$ has a minimum in this region. It is not difficult to see the reason for this phenomenon, by comparing the prior distribution of θ with the posteriors for various values of $\bar{x}$, as done in Figure C2 for $\bar{x} = 500$; 3622; and 8000. We observe the following:

Observations near the prior expectation: Confidence in the prior estimate is maximally increased, or confusion is maximally reduced, and thus $I_{\bar{x}}$ is (close to) maximum. On the other hand the posterior differs very little from the prior; the sample has reinforced the prior, but has added little new information, and $J_{\bar{x}}$ is (close to) minimum.

Observations unexpectedly far from the prior expectation: The spread of the posterior is increased, i.e. confidence in estimates of θ is reduced, leading to a negative value of $I_{\bar{x}}$. On the other hand, the sample data has demonstrated that certain values for θ which were a priori plausible at least, are in fact very unlikely (viz. values of θ greater than 6000 when $\bar{x} = 500$, or values less than 1500 when $\bar{x} = 8000$); the data have contributed distinctly *new* information relative to the prior, and $J_{\bar{x}}$ is in consequence large.

It seems that it may be of interest to compare $I_{\bar{x}}$ and $J_{\bar{x}}$ in the various other cases considered by the authors. Certainly these two quantities provide complementary measures of the effect of sample data on the information state relative to the prior. In terms of the title of the authors' paper, however, I view $J_{\bar{x}}$, the Kullback-Leibler measure, as giving the more meaningful measure of the new information content of the data.

Figure C1

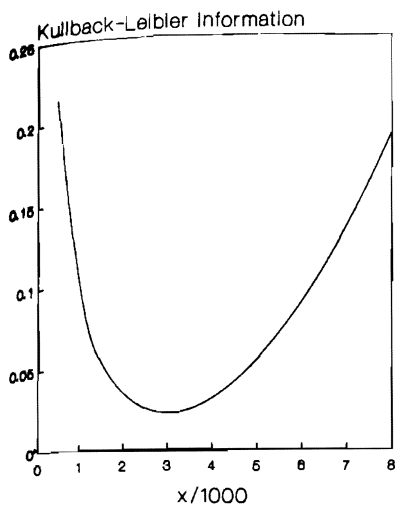


Figure C2

